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RESEARCH

Effect of sub-minimum inhibitory concentration of ceftriaxone on the expression of outer membrane proteins in *Salmonella enterica* serovar Typhi

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Abstract

Multi-drug resistance (MDR) in *Salmonella* is one of the major reasons for foodborne outbreaks worldwide. Decreased susceptibility of *Salmonella* Typhi to first-line drugs such as ceftriaxone, ciprofloxacin, and azithromycin has raised concern. Reduced outer membrane proteins (OMPs) permeability and increased efflux pump transportation are considered to be the main reasons for the emergence of antibiotic resistance in *Salmonella.* The present study aimed to assess the expression of OMPs at sub-lethal concentrations of ceftriaxone in *S*. Typhi (Sl5037/BC, and Sl05). The *S*. Typhi strains were exposed to sub-MIC and half of the sub-MIC concentrations of ceftriaxone at three different time intervals (0 min, 40 min, and 180 min) and analyzed for differential expression of OMPs. Further, the expression variation of OMP encoding genes (*yaeT*, *ompX*, *lamb*, *ompA*, and *ybfM*) in response to ceftriaxone was evaluated using real-time PCR. The genes like *lamB*, *ompX*, and *yaeT* showed significant downregulation ($p < 0.05$) compared to the control without antibiotic exposure, whereas *ybfM* and *ompA* showed a moderate downregulation. The expression of *omp* genes such as *lamB*, *ompA*, *ompX*, *ybfM*, and *yaeT* were found to be low in the presence of ceftriaxone, followed by time and dose-dependent. The study provides insights into the possible involvement of OMPs in drug resistance of *S*. Typhi, which could help develop a therapeutic strategy to combat MDR isolates of *S*. Typhi.

Keywords *Salmonella* Typhi · Antimicrobial resistance · Ceftriaxone · Outer membrane proteins · OMPs

Introduction

Salmonella spp. is one of the major causative agents of food and waterborne illness across the world. In the Southeast Asian region, foodborne illnesses are responsible for an estimated more than 150 million cases, and 175,000 deaths yearly, and nearly one-third (30%) of all deaths are among children under the age of 5 years (WHO 2016). Therefore, the pathogen has been considered one of the priority pathogens on the WHO list for future research and antimicrobial drug development strategies (WHO 2017; FDA 2019). The pathogen is classified into two species, *Salmonella enterica,* and *Salmonella bongori*, where *Salmonella enterica* is split into six subspecies with more than 2600 serotypes. The majority of enteric illness is caused by *S. enterica* subsp. *enterica*, which has more than 2500 serovars (Xu et al. 2021). Generally, *Salmonella* is transmitted to humans through the consumption of contaminated food, egg, meat, poultry, and milk, and also spreads through contaminated water, a major problem in low and middle-income countries. The organism is found in human sewage, farm effluents, and any material with fecal contamination (Asfaw Ali et al. 2020). *Salmonella enterica* serovar Typhi is considered to be the causative agent of typhoid fever and can transmit through community-acquired systemic infection that remains a major public health problem in underdeveloped countries (John et al. 2016).

Chloramphenicol, ampicillin, co-trimoxazole, third-generation cephalosporins, and fluoroquinolones were the drugs of choice to treat the infection that can reduce typhoid fever mortality significantly. However, *Salmonella* acquired

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resistance to all three first-line antimicrobials in the 1980s, followed by fluoroquinolone resistance in the 1990s (Shetty et al. 2012). The emergence of fluoroquinolones and cephalosporin resistance is a major setback to the treatment of the infection. In addition, due to the high toxicity, the fourth-generation fluoroquinolones (gatifloxacin) were banned by the Government of India in 2011 (Koirala et al. 2013). The limited antibiotic prescription has restricted the usage of moxifloxacin in the treatment of typhoid fever (MHFW 2016). Many factors contribute to drug resistance in *Salmonella*, such as the acquisition of a plasmid or chromosomal mutation, reduced permeability of the antimicrobial agent, modification of the antimicrobial target site, and inactivation of the antimicrobial agent (Ugboko and De 2014; Deekshit et al. 2015). In addition to all these factors, the outer membrane proteins (OMPs) of Gram-negative bacteria are also known to play an essential role in drug resistance (Ghai and Ghai 2018; Maiti et al. 2021). The OMPs are porin proteins that act as a water-filled channel to allow the movement of specific molecules through the channel (Vergalli et al. 2020). Most of the hydrophilic antibiotics are diffused through the porin to gain access to the cell interior (Prajapati et al. 2021). Passive diffusion porins OmpA, OmpX, and OmpF, are the key OMPs of *S.* Typhi for the passage of the first line of drugs such as ceftriaxone and ciprofloxacin (Correia et al. 2017). The substrate-specific lambda membrane protein (LamB) is the predominant porins for the emergence of multidrug resistance in Gram-negative organisms (Sun et al. 2021). By altering OMPs, bacteria can acquire resistance to antimicrobial agents by reducing their permeability (Yang et al. 2020). However, studies to understand the function of various OMPs in antimicrobial resistance in *S.* Typhi are limited. In this study, we determined the involvement of multiple OMPs in promoting antibiotic resistance in *S.* Typhi.

Materials and methods

Bacterial culture

Thirty-eight isolates of *Salmonella enterica* serovar Typhi were retrieved from the institutional repository (Nitte University Centre for Science Education and Research, Mangalore) and inoculated into brain–heart infusion broth (BHI) (HiMedia, India). These isolates were initially isolated and kindly provided by Dr. Anusha Rohit, Department of Microbiology, The Madras Medical Mission, Chennai, India. The details of the isolates are provided in Table 1. The cultures were incubated at 37 °C for 24 h in a shaking incubator. The cultures were streaked onto xylose lysine deoxycholate (XLD) agar (HiMedia) and incubated at 37 °C for 24 h. The isolates observed as pink colonies with black centers after 24 h of incubation were considered *S.* Typhi. All the positive isolates were further confirmed by PCR targeting the *invA* gene of *Salmonella* (Adesiji et al. 2014).

Antibiotic susceptibility test

The disc diffusion test was done on Muller-Hinton agar (MHA) (HiMedia) as per the guidelines of the clinical and laboratory standards institute (CLSI 2019) for all the isolates. The isolates with the turbidity of 0.5 McFarland unit were swabbed on MHA plates and tested with antibiotics such as ampicillin (10 µg), amoxicillin (30 µg), cefotaxime (30 µg), ceftazidime (30 µg), aztreonam (30 µg), azithromycin (15 µg), meropenem (10 µg), nalidixic acid (30 µg), ciprofloxacin (5 µg), ceftriaxone (30 µg), cotrimoxazole (25 µg), tetracycline (30 µg), chloramphenicol (30 µg), cefoperazone (75 µg). These antibiotics were dispensed on the agar surface with forceps and incubated for 16–18 h at 37 °C. The isolates were classified as sensitive, intermediate, and resistant based on the susceptibility test results. Four sensitive (Sl05, Sl5037/BC, Sl28, and Sl3939) and resistant isolates of *S.* Typhi (Sl07, Sl02, Sl15, and Sl13984) were selected for further studies (Table 1). A *Salmonella enterica* subsp. *enterica* ser. Typhimurium ATCC 14028 was used as a reference strain.

Determination of antibiotic resistance genes

Genomic DNA

The genomic DNA was extracted from four sensitive and four resistant isolates of *S.* Typhi according to the standard protocol (Aboul-Maaty and Oraby 2019) with minor

Table 1 List of clinical *Salmonella enterica* serovar Typhi isolates used for this study

Isolates IDs	Antibiotic sensitive isolates	Antibiotic-resistant isolates
Sl07, Sl27, Sl02, Sl15, Sl13984, Sl13865, Sl13605,Sl5037/BC, Sl3939, Sl28, Sl15059, Sl6149, Sl13417, Sl07, Sl14837, Sl14608, Sl08, Sl5917, Sl13865 Sl13554, Sl13966, Sl8132, Sl23, Sl30, Sl29, Sl18, Sl01, Sl19, Sl10, Sl04, Sl14, Sl15059, Sl6756, Sl8799, Sl9553, Sl03, Sl06, Sl14028[#]	Sl05* Sl5037/BC* Sl28 Sl3939	Sl07 Sl02 Sl15 Sl13984

*These two isolates were considered for differential expression studies of OMPs in the presence of ceftriaxone

[#]Reference strain of *Salmonella enterica* subsp. *enterica* ser. Typhimurium ATCC 14028 modifications. The extracted DNA pellet was resuspended in Tris–EDTA (TE) buffer. The concentrations and purity of the DNA samples were analyzed spectrophotometrically (Implen, Germany) and stored at − 20 °C for further use.

Plasmid DNA

The plasmid DNA was extracted from four sensitive and four resistant isolates of *S*. Typhi using a kit (Qiagen, Germany). Extracted plasmids were electrophoresed and visualized under UV gel documentation system (Bio-Rad, CA, USA).

Determination of antibiotic resistance gene

Antibiotic resistance genes such as ESBL genes ($bla_{\text{CTX-M}}$, bla_{TEM}, bla_{VIM}, bla_{OXA}), tetracycline resistance genes (*tet*A *tet*B *tet*C *tet*D *tet*E and *tet*G), sulfonamide resistance genes (*sul*1 and *sul*2), chloramphenicol resistance genes (*cat*B3), macrolide resistance gene (*mph*A), florphenicol resistance gene (*flo*R) were screened for the presence in selected four sensitive and resistant isolates by PCR assay. The primer details of the genes are summarized in Table 2. The PCR amplicons were subjected to agarose gel electrophoresis, stained with ethidium bromide (0.5 µg/ml), and visualized using a gel documentation system (Bio-Rad).

Extraction and profiling of OMPs of *S.* Typhi

OMP profiling

To extract total OMPs, four resistant and sensitive isolates of *S*. Typhi were grown in BHI broth at 37 °C overnight. The total OMPs were extracted as described previously by Maiti et al. (2011) with some modifications. Cells were harvested by centrifugation at 12,000×*g* for 10 min at 4 °C. Cell pellets were re-suspended in 10 mM tris HCl and disrupted by sonication at 20% amplitude with 10 s intervals for 7 min. Cell debris was removed by centrifugation, and total OMPs fraction was collected by centrifugation at 19,357×*g* at 4 °Cfor 6 h. The OMP pellets were mixed with 20% sarcosylamine (Sigma-Aldrich) and incubated overnight at 4 °C. Finally, total OMPs were collected by centrifuging the mixture at 20,817×*g* at 4 °C for 8 h and dissolved in 100 µl of sterile distilled water for further use.

The concentration of extracted OMPs was estimated by bicinchoninic acid (BCA) protein assay kit (Merck, USA) as per the manufacturer's guidelines. The extracted total OMPs were separated on 12% sodium dodecyl sulphate–polyacrylamide gel electrophoresis (SDS-PAGE) as per the method described previously (Brunelle and Green 2014) with minor modification using a vertical gel apparatus (Bio-Rad). The OMPs were then visualized using coomassie brilliant blue stain R-250 (HiMedia, India).

Whole-cell protein (WCP) profiling

For WCP profiling, four resistant and sensitive isolates of *S*. Typhi were grown in BHI broth at 37 °C for overnight. The cultures were centrifuged, and pellets were mixed with sample buffer. The cell lysate was incubated at 95 °C for 15 min, followed by incubation at room temperature for 15 min. Later, electrophoresis was carried out using 12% SDS-PAGE, followed by coomassie brilliant blue staining.

Expression of OMPs in the presence of ceftriaxone

Minimum inhibitory concentration (MIC) and experimental setup

Two *S*. Typhi isolates (Sl05, and Sl5037/BC), sensitive to ceftriaxone and other beta-lactamase drugs, were chosen for this study. The MIC was determined using Ezy MIC strips (HiMedia) by placing the strips aseptically onto MHA plates swabbed with young cultures and incubated for overnight. The MIC was determined where the eclipse intersects the MIC scale on the strip (Fig. 1). Two antibiotic doses were selected based on the MIC value, i.e., below MIC concentration and half of the below MIC concentration, and marked as dose:1 (DS1) and dose:2 (DS2), respectively, for further study. A control group was included without the addition of any antibiotic. Samples were taken at 3 time points (0, 40, and 180 min), centrifuged, and the pellet was used for both total RNA and total OMP extractions.

OMP extraction and profiling

The detailed procedure of OMP extraction and SDS-PAGE analysis have been described before.

Differential expression of gene encoding OMPs

The various OMP encoding genes were retrieved from the National Center for Biotechnology Information (NCBI) and screened using silico analysis. Selected genes were considered for the primer design. The details of primers have been listed in Table 2. Total RNA was isolated from samples at various time points using TRIzol reagent (Invitrogen, USA), and the concentration was determined spectrophotometrically (Implen, Germany). The quantitative real-time PCR (qPCR) analysis was carried out using Takara One-Step TB Green® PrimeScript™ RT-PCR Kit II (Takara, Japan) as per the manufacturer's protocol. The reactions were performed in a real-time detection system (CFX96 Touch™, BioRad, USA) with recommended cycling conditions. Additionally, a melting curve analysis was included. The data was recorded from BioRad CFX Manager (version 3.1) at the end of the assay. The relative gene expression analysis was carried out

Table 2 List of the primer used for this study

Gene name	Sequence (5′–3′)	Source/references	Purpose
yaeT	F: AGTAACACCATTCGCGCTCT R: TGGTCGGACGTTCTTTTACC	This study	qPCR assay
ompX	F: ACTGTTACCGGTGGTTACGC R: AGATCGTACTAACGGTGCGG		
lamB	F: TCCGTATTTTCGCCACCTAC R: TGGAAATGCCGGAGTTAGTC		
ompA	F: CAGGTTAACCCGTACGTTGG R: ACTGAACGCCCTGAGCTTTA		
ybfM	F: TTACGCCTGGGATGCTAAAC R: GGGTATAGACCGCATCCAGA		
bla_{VIM}	F: GTTTGGTCGCATATCGCAAC R: AATGCGCAGCACCAGGATAG	Fallah et al. (2013)	Detection of antibiotic determinants
bla_{TEM}	F: AGGAAGAGTATGATTCAACA R: CTCGTCGTTTGGTATGGC	Shi et al. (2013)	
blaoxa	F: TTGGTGGCATCGATTATCGG R: GAGCACTTCTTTTGTGATGGC	Poirel et al. (2012)	
$bla_{CTX\text{-}M}$	F: ACGTTAAACACCGCCATTCC R: TCGGTGACGATTTTAGCCGC	Colomer-Lluch et al. (2011)	
*sul*1	F: CGCACCGGAAACATCGCTGCAC R: TGAAGTTCCGCCGCAAGGCTCG	Ma et al. (2011)	
*sul*2	F: TCCGGTGGAGGCCGGTATCTGG R: CGGGAATGCCATCTGCCTTGAG	Ma et al. (2011)	
*tet*A	F: GCGCGATCTGGTTCACTCG R: AGTCGACAGYRGCGCCGGC	Ma et al. (2011)	
*tet*B	F: AAAACTTATTATATTATAGTC R: TGGAGTATCAATAATATTCAC	Ma et al. (2011)	
*tet*C	F: GCGGGATATCGTCCATTCCG R: GCGTAGAGGATCCACAGGACG	Ma et al. (2011)	
*tet*E	F: GTTATTACGGGAGTTTGTTGG R: AATACAACACCCACACTACGC	Ma et al. (2011)	
*tet*G	F: GCAGAGCAGGTCGCTGG R: CCYGCAAGAGAAGCCAGAAG	Ma et al. (2011)	
qnr_{VC}	F: GAGYTKTATGGTTTAGAYCCTCG R: TGTTCYTGYTGCCACGARCA	Martineau et al. (2000)	
*cat*B3	F: TCAAAGGCAAGCTGCTTTCTGAGC R: TATTAGACGAGCACAGCATGGGCA	Soge et al. (2006)	
Mph(A)	F: GTGAGGAGGAGCTTCGCGAG R: TGCCGCAGGACTCGGAGGTC	Nguyen et al. (2009)	
*flo*R	F: GTCATTCCTCACCTTCATCCTAC R: GACACCAGCACTGCCATTG	Khan et al. (2011)	
*inv*A	F: GTGCCGGTTTTATCGTGACT R: ATCGCCAATCAGTCCTAACG	Adesiji et al. (2014)	Detection of *Salmonella*

by Livak'smethod (Livak and Schmittgen 2001) using 16S rDNA as a housekeeping gene.

Statistical analysis

The data acquired from the qPCR were tested for significant differences using two-way ANOVA. The differences were considered statistically significant at ($P < 0.05$).

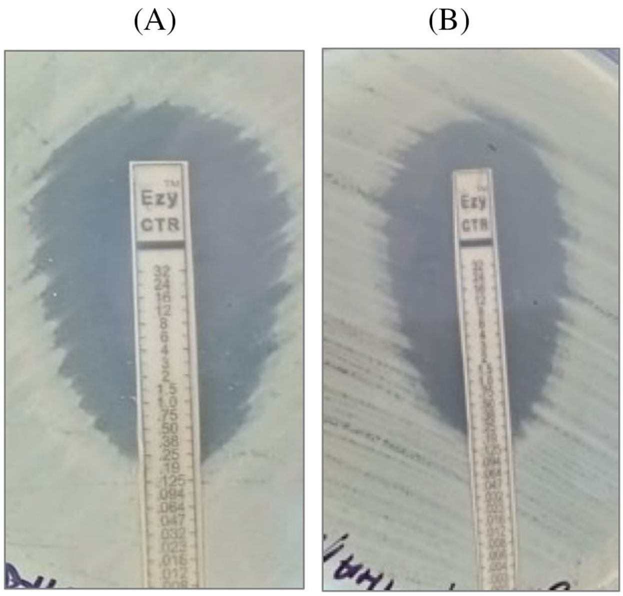

Fig. 1 The ceftriaxone E-test showing the inhibition against *S*. Typhi. The MIC concentration was determined as **A** 0.190 µg/ml (Sl5037/BC) and **B** 0.125 µg/ml (Sl05)

Results were analyzed using GraphPad Prism, version 8.4.3. (GraphPad, CA, USA).

Results

Bacterial culture and antibiotic sensitivity

The antibiotic sensitivity test of the isolates revealed high resistance to nalidixic acid (81.5%) and azithromycin (42.1%), while low resistance was seen towards ciprofloxacin (13.16%), cefotaxime (5.2%), cotrimoxazole (2.6%), ampicillin (2.6%), meropenem (2.6%), tetracycline (2.6%) and aztreonam (5.2%). However, antibiotics like amoxicillin, ceftazidime, cefoperazone, chloramphenicol, and ceftriaxone showed 100% sensitivity. Among the thirty-eight isolates, five showed multidrug resistance; one isolate showed resistance to six out of seventeen antimicrobial drugs tested.

Determination of antibiotic resistance gene

The antibiotic resistance determinants test of the eight isolates (four sensitive and four resistant) was carried out using genomic and plasmid DNA. All the isolates showed the presence of bla_{TEM}, $bla_{OXA,}$ and bla_{VIM} in the plasmid. Isolates Sl02 and Sl13984 were found to be positive for multiple antibiotic resistance genes. The 40% of isolates were positive for *CatB3* and bla_{CTX}. However, none of the isolates were positive for *tet*A, *tet*B, *tet*C, *tet*D, *tet*E and *tet*G, *sul*1, *sul*2 and *sul*3, *flo*R, and *mph*A. On the other hand, isolates tested using genomic DNA did not show the presence of any of the antibiotic resistance genes tested.

The OMP and WCP profiling of antibiotic-resistant and sensitive *S.* Typhi

In OMP profiling, intense bands were observed between the regions of 35 and 60 kDa (Fig. 2a), except for Sl3939 which showed low-intensity bands in SDS-PAGE. Lower intensity bands were also observed in Sl05, Sl02, Sl3939, Sl15, Sl5037/BC, and Sl13984 at a molecular weight of ~ 35 kDa. Additionally, few clear thick bands were also observed in all the isolates at ~ 35, ~ 45, ~ 60, and ~ 80 kDa. However, WCP profiling of sensitive and resistant isolates

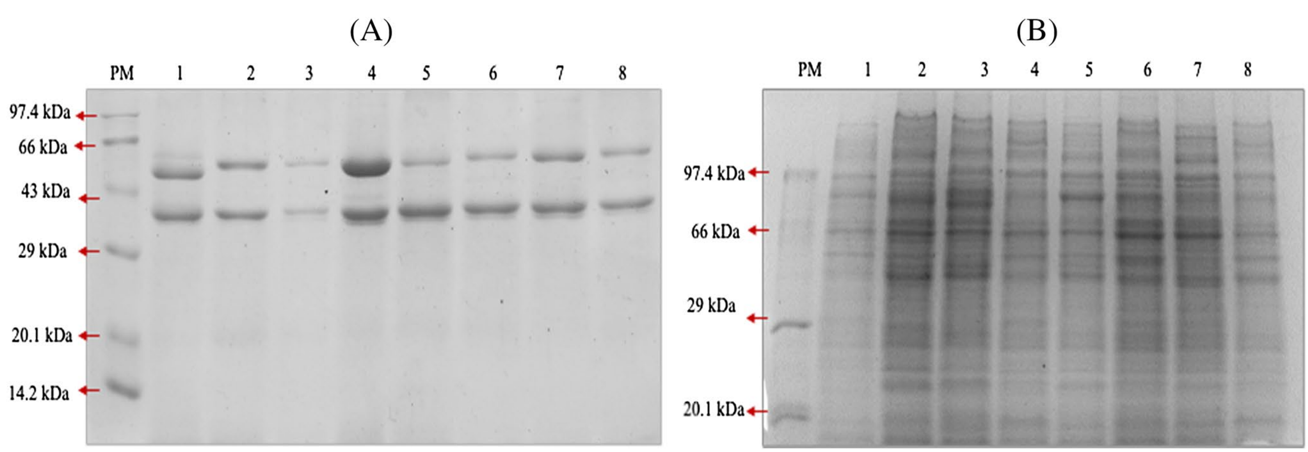

Fig. 2 A 12% SDS-PAGE gel analysis using coomassie brilliant blue staining method. **A** Gel showing OMP profile of *S*. Typhi isolates. PM-Protein molecular weight marker (Genei, India), Lane 1-Sl28, Lane 2-Sl07, Lane 3-Sl3939, Lane 4-Sl02, Lane 5-Sl05, Lane 6-Sl15, Lane 7-Sl5037/BC, Lane 8-Sl13605. **B** Gel showing the WCP profile of *S*. Typhi. PM-Protein molecular weight marker (Genei, India), Lane 1-Sl28, Lane 2-Sl07, Lane 3-Sl3939, Lane 4-Sl02, Lane 5-Sl05, Lane 6-Sl15, Lane 7-Sl5037/BC, Lane 8-Sl13984

of *S*. Typhi yielded several polypeptide bands ranging from 20 to 100 kDa (Fig. 2b).

The MIC of ceftriaxone and exposed dose

The MIC concentration of ceftriaxone was determined as 0.190 µg/ml, and 0.125 µg/ml for Sl5037/BC and Sl05 isolate, respectively (Fig. 1). Based on MIC, the sub-MIC concentration (DS1) of Sl5037/BC was 0.125 µg/ml, and half of the sub-MIC concentration (DS2) was 0.06 µg/ml. Similarly, the DS1 and DS2 of Sl05 were 0.094 and 0.04 µg/ml, respectively.

Effect of ceftriaxone on OMP profiling

Total OMPs were extracted from the ceftriaxone exposed isolates (Sl05 and Sl5037/BC) and from the control group (without ceftriaxone) and analyzed by 12% SDS-PAGE analysis. The gel showed about two to five OMP polypeptide bands, ranging from ~32 to ~60 kDa. An intense band was observed at ~35 kDa in all the samples. (Fig. 3).

Differential expression of *omp* genes in the presence of ceftriaxone

The expression of different genes encoding OMPs of these two isolates sensitive to ceftriaxone was analyzed by qPCR assay (Fig. 4). However, all the *omp* genes used in this study showed downregulation. A significant downregulation was observed in *lamB* at 40 min (4.03-fold) and 180 min (5.1-fold) for DS1, and at 40 min (4.2-fold) and 180 min (4.4-fold) for DS2 (Fig. 4A). The effect of ceftriaxone downregulated *ompA* as compared to control at 40 min (1.1-fold) and 180 min (onefold) for DS1, and at 40 min (0.4-fold) and 180 min (1.7-fold) for DS2 (Fig. 4B). Similarly, *ompX* was downregulated at 40 min (1.7-fold) and 180 min (3.1-fold) for DS1, and at 40 min (1.2-fold) and 180 min (3.2-fold) for DS2 (Fig. 4C). In the case of *yaeT* gene, the expression was downregulated at 40 min (3.9-fold) and 180 min (7.7-fold) for DS1, and at 40 min (4.5-fold) and 180 min (6.6-fold) for DS2 (Fig. 4D). Whereas, *ybfM* gene was also downregulated at 40 min (3.6-fold) and 180 min (4.5-fold) for DS1, and at 40 min (2.9-fold) and 180 min (3.1-fold) for DS2 (Fig. 4E).

Discussion

Foodborne illnesses caused by *Salmonella,* along with *Campylobacter* and *Clostridium perfringens,* are increasing worldwide with the increase in the emergence of MDR and extensively drug-resistant strains (CDC, 2019). Horizontal gene transfer is the primary reason for the widespread dissemination of resistance in distinct bacterial species (Silbergeld et al. 2008). The random use of antibiotics

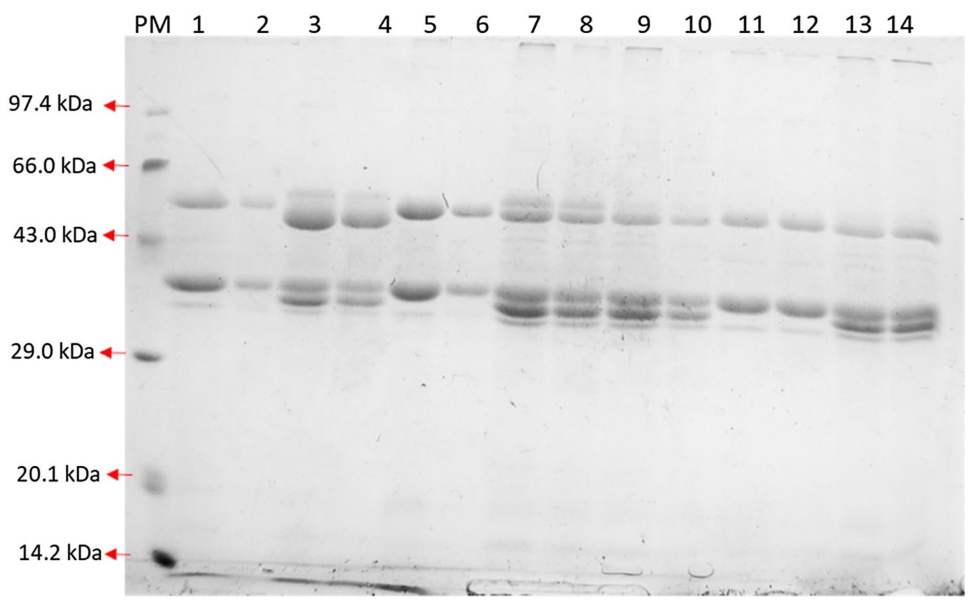

Fig. 3 The12% SDS-PAGE gel showing OMP profile of *S*. Typhi. PM-Protein molecular weight marker (Genei, India). Two antibiotic-sensitive isolates, Sl5037/BC and Sl05 were exposed to ceftriaxone and used for this study. Same isolates without ceftriaxone exposure were considered as control. DS1: below MIC concentration and DS2: half of the below MIC concentration. Lane 1: 0 min-control (Sl5037/BC), Lane 2: 0 min-control (Sl05), Lane 3: 40 min-control (Sl5037/BC), Lane 4: 40 min-control (Sl05), Lane 5: 40 min-DS1 (Sl5037/BC), Lane 6: 40 min-DS1 (Sl05), Lane 7: 40 min-DS2 (Sl5037/BC), Lane 8: 40 min-DS2 (Sl05), Lane 9: 180 min-control (Sl5037/BC), Lane 10: 180 min-control (Sl05), Lane 11: 180 min-DS1 (Sl5037/BC), Lane 12: 180 min-DS1 (Sl05), Lane 13: 180 min-DS2 (Sl5037/BC), Lane 14: 180 min-DS2 (Sl05)

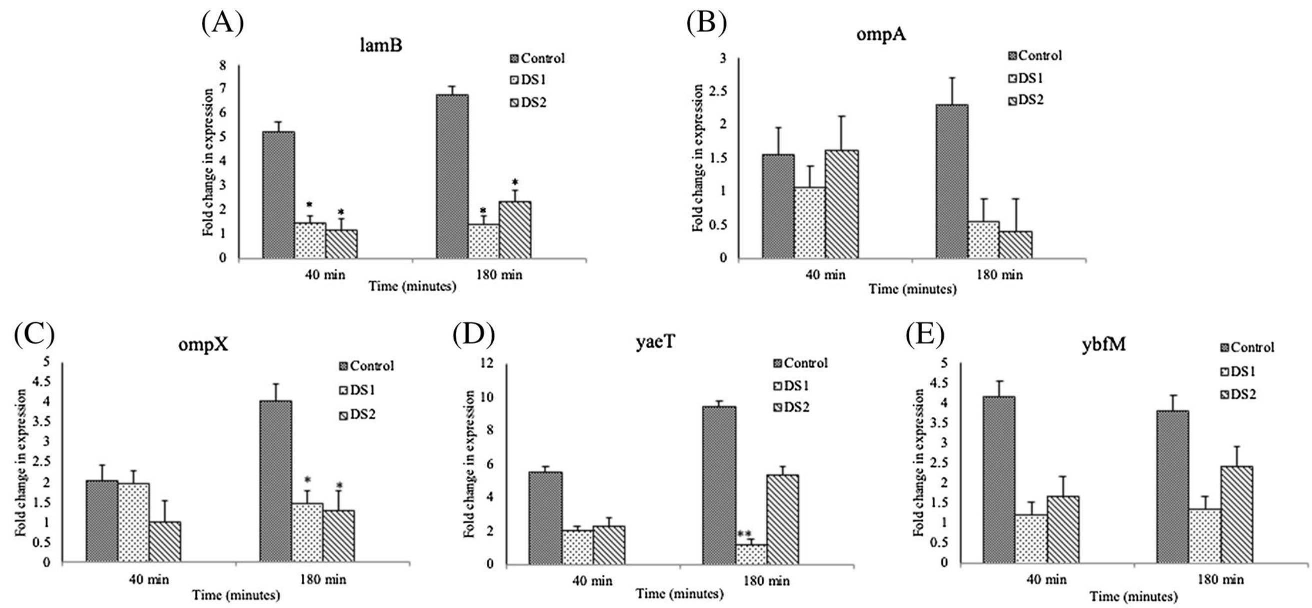

Fig. 4 Differential expression of OMP encoding genes using qPCR assay. Two antibiotic-sensitive isolates and used for this study. Two isolates (Sl5037/BC and Sl05) were exposed to two different doses (DS1: below MIC concentration and DS2: half of the below MIC concentration) of ceftriaxone. Isolates without ceftriaxone exposure were considered as control. **A** *lamB*, **B** *ompA*, **C** *ompX*, **D** *yaeT*, and **E** *ybfM*. P value are: <0.001 for (***) very significant, 0.001 to 0.01 for (**) very significant, 0.01 to 0.05 for (*) for significant and ≥ 0.05 for non-significant values

has resulted in the emergence of multi-drug resistance and extensive drug resistance isolates of *S*. Typhi in developing nations (Akram et al. 2020; Ugboko and De 2014). The surge of resistance to cephalosporins and quinolones in *S*. Typhi is due to various mechanisms; one such mechanisms are loss of permeability in OMP and increase in the expression of efflux pump. Park et al. (2011) reported that OmpA is one of the most abundant OMP in Gram-negative bacteria that might contribute to drug resistance. The 28–36 kDa OmpA is one of the best characterized OMPs in Enterobacterales family (Nie et al. 2020). *S*. Typhi is also known to produce β-lactamase enzymes that lead to the resistance towards various drugs, including the third generation of cephalosporins (Threlfall 2013). Batchelor et al. (2005) reported the bla_{CTXM} mediated resistance is due to the horizontal transfer of plasmid in the bacteria. Ahmed et al. (2014) also reported a rare phenomenon of multi-ESBL producing *S*. Typhi harboring bla_{CTXM} and bla_{TEM}. However, in the present study, the bla_{TEM}, bla_{VIM}, bla_{OXA} were the most encountered β-lactamase genes in all the tested isolates. In contrast, bla_{CTXM} and *catB3* were found in 50% (4/8) of the isolate. Similar findings have been reported by Kpoda et al. (2018), where ESBL encoding genes such as bla_{TEM} and bla_{CTXM} were present in most of the *E. coli* isolates. Whereas, Dembélé et al. (2020) demonstrated the prevalence of the bla_{OXA} as one of the important β-lactam producing genes in *Salmonella.* This suggests that the occurrence of β-lactamase gene is highly variable in β-lactam resistance *Salmonella* isolates. This is possible considering the fact that extended-spectrum cephalosporins and other β-lactam antibiotic resistance have been conferred, primarily due to the horizontal gene transfer. The genes *qnrVC*, *tet*, *sul*, and *mphA* were not detected in the current study. In contrast, the occurrence of *qnr* (quinoline) and *bla* (β-lactamase) has been associated with multi-drug resistance in *Enterobacterales* (Metuor-dabire et al. 2019). The easy transmission of the plasmid-mediated quinoline resistance genes along with the cephalosporin resistance gene results in the dissemination of multidrug resistance in communities and hospitals. Next, we examined the response of OMPs in antibiotic-sensitive and resistant isolates of *S*. Typhi by OMP profiling through SDS-PAGE analysis. The previous study has shown the key role of OMPs in the emergence of MDR and XDR due to the diffusion of molecules inside the cell of *S*. Typhi (Choi and Lee 2019).

It has also been reported that cefoxitin and carbapenem resistance isolates of *Klebsiella pneumoniae* showed the loss of OmpK36 in the region of 40 kDa (Shi et al. 2013). Abd El-Baky et al. (2020) also reported that colistin resistance and sensitive isolates show the difference in the expression of OMPs in SDS-PAGE analysis. Our results also showed variation in the expression of OMPs in antibiotic-resistant and sensitive isolates of *S*. Typhi. Eight selected isolates showed the presence of major OMPs at regions of 35–60 kDa in SDS-PAGE analysis. This is possibly due to the change in expression incited by resistance activity towards the antibiotics. A similar result was also observed by Maiti et al. (2011), wherein OMP analysis of *Edwardsiella tarda* showed the presence of major bands with the variation in expression at the same region between 36 and 50 kDa. Similarly, a study by Harikrishnan et al. (2013) also demonstrated the variation of OMPs expression in *Shigella flexneri* clinical isolates between the 29 and 57 kDa region.

Several studies have reported that sub-MICs of antibiotics can significantly increase the doubling time with a reducing bacterial growth rate. In the study, we used ceftriaxone to analyze the differential expression of *omp* genes in two *S*. Typhi isolates (SL5037/BC and Sl05) using qPCR assay. Since ceftriaxone is a drug of choice for typhoid fever for clinicians after cefixime, it has a major role in the emergence of MDR in *S*. Typhi (Dahiya et al. 2019). In the exposure to ceftriaxone, all the genes encoding OMPs (*lamB*, *ompA*, *ompX*, *yaeT*, and *ybfM*) showed downregulation compared to control (without ceftriaxone). The downregulation of porins leads to a surge in resistance to carbapenem class of drugs; similarly, for *β*-lactam class of drugs such as ceftriaxone, such phenomenon reduces the uptake of antimicrobial drugs, strictly restricts the intracellular drug accession that leads towards the emergence of resistance via mutational changes or enzymatic changes in it (Prajapati et al. 2021).

We extracted OMPs by sarcosylamine method from ceftriaxone exposed isolates of *S*. Typhi (Sl05 and SL5037/BC). Our result revealed an intense band at ~ 35 kDa in all the samples in both doses (DS1-DS2). However, there was not much variation in OMP bands. Hoštacká and Karelová (1997) have also reported no change in the OMPs band of *Pseudomonas aeruginosa* after the treatment of different concentrations of amikacin. Maiti et al. (2011) previously reported the presence of OMP bands at ~ 35 kDa with a minor change in intensity of bands.

Der (2016) stated that during oxidative stress, the OmpA pore is actively responsible for opening or closing of the pore channel. Dam et al. (2018) reported that the expression of OmpA in RNA levels can decrease due to stress response toward antibiotics in less than 60 min. Similarly, in our study, 180 min post-exposure to ceftriaxone, the *ompA* has been slightly downregulated in both doses. Viveiros et al. (2007) noted that downregulation of *ompA* might lead to decrease in the uptake of hydrophilic antimicrobial drugs that tends toward the developing MDR. Also, another important general diffusion porin, OmpX, is closely associated with OmpA. OmpX permits several hydrophilic antimicrobial classes of drugs, including fluoroquinolones and *β*-lactams, for the passage (Dupont et al. 2007). Studies have indicated that downregulation of OmpX facilitates reduction of the cell uptake with the increase in cell invasion (Gupta et al. 2021). The ompX gene was downregulated in our study ($p < 0.05$). This downregulation of *ompX* might be due to the sub-lethal concentration of ceftriaxone that exhibited the stress in the porins, resulted in a decrease in the porin permeability, and has pushed towards the emergence of MDR in *Salmonella*. Similar investigation by Dupont et al. (2007) reported the downregulation of *ompX* during antimicrobial stress, which can decrease the porin permeability use by fluoroquinolone and β-lactam class of drug in Gram,-negative bacteria, leading to a resistance in pathogens. These results suggest that both *ompA* and *ompX* can play a key role in the development of MDR. LamB, another key porin that is specific for the diffusion of maltose and maltodextrin in the cell, has significant participation in acquiring resistance in β-lactams and fluoroquinolones. The studies of Lin et al. (2014) reported the downregulation of *lamB* after the exposure to MIC of chloramphenicol in *E. coli* isolates. In our study also, *lamB* expression was found to be significantly ($p < 0.05$) downregulated in both the concentration (DS1 and DS2) of ceftriaxone. We speculate this decrease the expression is associated with reducing the uptake of sugar in the bacterial cell due to environmental stress by ceftriaxone, which pushes toward reduction in the influx of drugs that surge towards resistance in *Salmonella*. Another study by Lin et al. (2010) commented that sub-MIC exposure of chlortetracycline in *E. coli* can exhibit negative regulation in porins. However, this finding also suggests that the different classes of antibiotics such as ceftriaxone, chlortetracycline, chloramphenicol, or other β-lactam drugs altogether have a direct relationship with the sugar-specific porin channel (LamB) by reducing the porin channel for the emergence of drug resistance. We also studied *yaeT*, another important OMP encoding gene. YaeT or Omp85, which is less studied in Gram-negative organisms, is also identical to tob55 protein seen in plant and animal mitochondria (Malinverni and Silhavy 2007). This protein is essential for the early steps of outer membrane assembly and insertion of beta-barrel protein in the OMPs (Jain and Goldberg 2007). However, not much is known about its role in antimicrobial resistance. Nevertheless, in our study, the *yaeT* expression was significantly downregulated ($p < 0.05$), possibly due to the external stress response by ceftriaxone might reduce the putative binding site of *yaeT* and also reduce the OMP biogenesis. Sousa (2019) have reported that extra stress given by the antibiotic can reduce the binding of envelope protein for the influx of antimicrobial drugs. YbfM, which has been less studied in *Salmonella,* and not much known about its role in drug resistance. Tambe et al. (2006) noted that YbfM has a structural similarity with OprD in *Pseudomonas*. Our study showed the downregulation of *ybfM* expression. However, further study requires to reveal the role of *ybfM* in decreased expression in the exposure to β-lactam drugs.

Conclusion

In conclusion, the differential expression of five genes encoding OMPs (*lamB*, *ompA*, *ompX*, *ybfM*, and *yaeT*) showed reduced expression in general, but their degree of expression was time and dose-dependent. Whereas OMP profiles below MIC and sub-MIC of ceftriaxone against time did not exhibit any significant changes in the banding pattern of OMPs in the native SDS-PAGE analysis. Nevertheless, further studies are required to delineate the exact role of these OMPs in antibiotic resistance in *S.* Typhi.

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Author contributions SDA performed experiments, interpreted the results, analyzed the data, and wrote the manuscript. KPA performed experiments. VKD designed the experiments, supervised, and reviewed the manuscript. AR provided *Salmonella* isolates, supervised, and reviewed the manuscript. BM conceptualized, designed the experiments, supervised, provided funds and resources, and reviewed the manuscript.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

References

Abd El-Baky R, Masoud S, Mohamed D, Waly N, Shafik E, Mohareb D et al (2020) Prevalence and some possible mechanisms of colistin resistance among multidrug resistant and extensively drug-resistant *Pseudomonas aeruginosa*. Infect Drug Resist 13:323–332. https://doi.org/10.2147/idr.s238811

Aboul-Maaty NAF, Oraby HAS (2019) Extraction of high-quality genomic DNA from different plant orders applying a modified CTAB-based method. Bull Natl Res Cent 43:25. https://doi.org/10.1186/s42269-019-0066-1

Adesiji YO, Deekshit VK, Karunasagar I (2014) Antimicrobial-resistant genes associated with *Salmonella* spp. isolated from human, poultry, and seafood sources. Food Sci Nutr 2:436–442. https://doi.org/10.1002/fsn3.119

Ahmed D, Ud-Din AIMS, Wahid SUH, Mazumder R, Nahar K, Hossain A (2014) Emergence of bla_{TEM} Type Extended-Spectrum β-Lactamase Producing *Salmonella* spp. in the Urban Area of Bangladesh. ISRN Microbiol 2014:1–3. https://doi.org/10.1155/2014/715310

Akram J, Khan AS, Khan HA, Gilani SA, Akram SJ, Ahmad FJ, Mehboob R (2020) Extensively drug-resistant (XDR) typhoid: evolution, prevention, and its management. Biomed Res Int 2020:1–7. https://doi.org/10.1155/2020/6432580

Asfaw Ali D, Tadesse B, Ebabu A (2020) Prevalence and antibiotic resistance pattern of *Salmonella* isolated from caecal contents of exotic chicken in Debre Zeit and Modjo, Ethiopia. Int J Microbiol 2020:1–6. https://doi.org/10.1155/2020/1910630

Batchelor M, Hopkins K, Threlfall EJ, Clifton-Hadley FA, Stallwood AD, Davies RH, Liebana E (2005) $bla_{\mathrm{CTX\text{-}M}}$ genes in clinical *Salmonella* isolates recovered from humans in England and Wales from 1992 to 2003. Antimicrob Agents Chemother 49:1319–1322. https://doi.org/10.1128/AAC.49.4.1319-1322.2005

Brunelle JL, Green R (2014) One-dimensional SDS-polyacrylamide gel electrophoresis (1D SDS-PAGE). Meth Enzymol 541:151–159. https://doi.org/10.1016/b978-0-12-420119-4

CDC (2019) Antibiotic resistance threats in the United States, 2019 (2019 AR threats report). Centers for disease control and prevention. https://www.cdc.gov/drugresistance/biggest-threats.html. Accessed on 26 June 2021

Choi U, Lee CR (2019) Distinct roles of outer membrane porins in antibiotic resistance and membrane integrity in *Escherichia coli*. Front Microbiol 10:953. https://doi.org/10.3389/fmicb.2019.00953

CLSI (2019) Performance standards for antibiotic susceptibility testing. Clinical and laboratory standards institute. https://clsi.org/standards/products/free-resources/access-our-free-resources/. Accessed on 12 March 2021

Colomer-Lluch M, Jofre J, Muniesa M (2011) Antibiotic resistance genes in the bacteriophage DNA fraction of environmental samples. PLoS ONE 6:17549. https://doi.org/10.1371/journal.pone.0017549

Correia S, Hébraud M, Chafsey I, Chambon C, Viala D, Sáenz et al (2017) Comparative subproteomic analysis of clinically acquired fluoroquinolone resistance and ciprofloxacin stress in *Salmonella* Typhimurium DT104B. Proteomics Clin Appl 11:7–8. https://doi.org/10.1002/prca.201600107

Dahiya S, Malik R, Sharma P, Sashi A, Lodha R, Kabra SK, Sood S, Das BK, Walia K, Ohri VC, Kapil A (2019) Current antibiotic use in the treatment of enteric fever in children. IJMR 149:263–269. https://doi.org/10.4103/ijmr.IJMR_199_18

Dam S, Pages JM, Masi M (2018) Stress responses, outer membrane permeability control and antimicrobial resistance in Enterobacteriaceae. Microbiology 164:260–267. https://doi.org/10.1099/mic.0.000613

Deekshit VK, Kumar B, Rai P, Karunasagar I (2015) Differential expression of virulence genes and role of *gyrA* mutations in quinolone resistant and susceptible strains of *Salmonella* Weltevreden and Newport isolated from seafood. J Appl Microbiol 119:970–80. https://doi.org/10.1111/jam.12924

Dembélé R, Konaté A, Traoré O, Kaboré WAD, Soulama I, Kagambèga A, Traoré AS, Guessennd NK, Aidara-Kane A, Gassama-Sow A, Barro N (2020) Extended spectrum beta-lactamase and fluoroquinolone resistance genes among *Escherichia coli* and *Salmonella* isolates from children with diarrhea, Burkina Faso. BMC Pediatr 20:1–9. https://doi.org/10.1186/s12887-020-02342-z

Der HJ (2016) Rapidly regulates membrane permeability to survive oxidative stress. ASM 7:1–11. https://doi.org/10.1128/mBio.01238-16

Dupont M, James CE, Chevalier J, Pagès JM (2007) An early response to environmental stress involves regulation of OmpX and OmpF, two enterobacterial outer membrane pore-forming proteins. Antimicrob Agents Chemother 51:3190–3198. https://doi.org/10.1128/AAC.01481-06

Fallah F, Borhan RS, Hashemi A (2013) Detection of bla(IMP) and bla(VIM) metallo-β-lactamases genes among *Pseudomonas aeruginosa* strains. Int J Burn Trauma 3:122–124

FDA (2019) NARMS Update: integrated report summary. the national antimicrobial resistance monitoring system. https://www.fda.gov/animal-veterinary/national-antimicrobial-resistance-monitoring-system/2019-narms-update-integrated-report-summary. Accessed on 26 July 2021

Ghai I, Ghai S (2018) Understanding antibiotic resistance via outer membrane permeability. Infect Drug Resist 11:523–530. https://doi.org/10.2147/idr.s156995

Gupta A, Jeyakumar E, Lawrence R (2021) Strategic approach of multifaceted antibacterial mechanism of limonene traced in *Escherichia coli*. Sci Rep 11:1–15. https://doi.org/10.1038/s41598-021-92843-3

Harikrishnan H, Ismail A, Banga Singh KK (2013) Temperature-regulated expression of outer membrane proteins in *Shigella flexneri*. Gut Pathog 5:38. https://doi.org/10.1186/1757-4749-5-38

Hoštacká A, Karelová E (1997) Outer membrane proteins and elastase of *Pseudomonas aeruginosa* after the postantibiotic effect induced by amikacin. Chemotherapy 43:118–122. https://doi.org/10.1159/000239545

Jain S, Goldberg MB (2007) Requirement for YaeT in the outer membrane assembly of autotransporter proteins. J Bacteriol 189:5393–5398. https://doi.org/10.1128/JB.00228-07

John J, Van Aart CJC, Grassly NC (2016) The burden of typhoid and paratyphoid in India: systematic review and meta-analysis. PLoS Negl Trop Dis 10:1–14. https://doi.org/10.1371/journal.pntd.0004616

Khan SA, Sung K, Nawaz MS (2011) Detection of *aacA-aphD*, *qacEδ1*, *marA*, *floR*, and *tetA* genes from multidrug-resistant bacteria: comparative analysis of real-time multiplex PCR assays using EvaGreen® and SYBR® Green I dyes. Mol Cell Probes 25:78–86. https://doi.org/10.1016/j.mcp.2011.01.004

Koirala S, Basnyat B, Arjyal A, Shilpakar O, Shrestha K, Shrestha R, Shrestha UM, Agrawal K et al (2013) Gatifloxacin versus ofloxacin for the treatment of uncomplicated enteric fever in Nepal: an open-label, randomized, controlled trial. PLoS Negl Trop Dis 7:2523. https://doi.org/10.1371/journal.pntd.0002523

Kpoda DS, Ajay A, Somda M, Traore O, Guessennd N, Ouattara AS, Sangare L, Traore AS, Dosso M (2018) Distribution of resistance genes encoding ESBLs in *Enterobacteriaceae* isolated from biological samples in health centers in Ouagadougou, Burkina Faso. BMC Res Notes 11:5–9. https://doi.org/10.1186/s13104-018-3581-5

Lin XM, Yang JN, Peng XX, Li H (2010) A novel negative regulation mechanism of bacterial outer membrane proteins in response to antibiotic resistance. J Proteome Res 9:5952–5959. https://doi.org/10.1021/pr100740w

Lin X, Yang M, Li H, Wang C, Peng XX (2014) Decreased expression of LamB and Odp1 complex is crucial for antibiotic resistance in *Escherichia coli*. J Proteomics 98:244–253. https://doi.org/10.1016/j.jprot.2013.12.024

Livak KJ, Schmittgen TD (2001) Analysis of relative gene expression data using real-time quantitative PCR and the $2^{-\Delta\Delta CT}$ method. Methods 25:402–408. https://doi.org/10.1006/meth.2001.1262

Ma Y, Wilson CA, Novak JT, Riffat R, Aynur S, Murthy S, Pruden A (2011) Effect of various sludge digestion conditions on sulfonamide, macrolide, and tetracycline resistance genes and class I integrons. Environ Sci Technol 45:7855–7861. https://doi.org/10.1021/es200827t

Maiti B, Shetty M, Shekar M, Karunasagar I, Karunasagar I (2011) Recombinant outer membrane protein A (OmpA) of *Edwardsiella tarda*, a potential vaccine candidate for fish, common carp. Microbiol Res 167:1–7. https://doi.org/10.1016/j.micres.2011.02.002

Maiti B, Chakraborty A, Karunasagar I (2021) Biotechnological advances in the development of outer membrane protein-based vaccines for use in aquaculture. In: Gupta S (ed) Biotechnological advances in aquaculture health management. Springer, Singapore, pp 43–59

Malinverni J, Silhavy TJ (2007) The identification of the YaeT complex and its role in the assembly of bacterial outer membrane *β*-barrel proteins. Enzymes 25:129–149. https://doi.org/10.1016/S1874-6047(07)25006-1

Martineau F, Picard FJ, Lansac N, Ménard C, Roy PH, Ouellette M, Bergeron MG (2000) Correlation between the resistance genotype determined by multiplex PCR assays and the antibiotic susceptibility patterns of *Staphylococcus aureus* and *Staphylococcus epidermidis*. Antimicrob Agents Chemother 44:231–238. https://doi.org/10.1128/AAC.44.2.231-238.2000

Metuor-Dabire A, Sougue S, Tiemtore R, Zohoncon T, Bangre Y, Ouedraogo P et al (2019) Coexistence between (TOHO-type and BES-type) extended-spectrum-lactamase genes of identified enterobacteria at Saint Camille Hospital, Ouagadougou, West Africa. Int J Genet Mol Biol 11(2):34–40. https://doi.org/10.5897/ijgmb2019.0181

MHFW (Ministry of Health & Family Welfare, Government of India) (2016) National treatment guidelines for antimicrobial use in infectious diseases. NCDC, GoI. https://ncdc.gov.in/showfile.php?lid=649. Accessed on 26 July 2021

Nguyen MCP, Woerther PL, Bouvet M, Andremont A, Leclercq R, Canu A (2009) *Escherichia coli* as reservoir for macrolide resistance genes. Emerg Infect Dis 15:1648–1650. https://doi.org/10.3201/eid1510.090696

Nie D, Hu Y, Chen Z, Li M, Hou Z, Luo X, Mao X, Xue X (2020) Outer membrane protein A (OmpA) as a potential therapeutic target for *Acinetobacter baumannii* infection. J Biomed Sci 27:1–8. https://doi.org/10.1186/s12929-020-0617-7

Park JS, Lee WC, Choi S, Yeo KJ, Song JH, Han YH, Lee JC, Kim SIl, Jeon YH, Cheong C, Kim HY, (2011) Overexpression, purification, crystallization and preliminary X-ray crystallographic analysis of the periplasmic domain of outer membrane protein A from *Acinetobacter baumannii*. Acta Crystallogr F 67:1531–1533. https://doi.org/10.1107/S1744309111038401

Poirel L, Potron A, Nordmann P (2012) OXA-48-like carbapenemases: the phantom menace. J Antimicrob Chemother 67:1597–1606. https://doi.org/10.1093/jac/dks121

Prajapati JD, Kleinekathöfer U, Winterhalter M (2021) How to enter a bacterium: bacterial porins and the permeation of antibiotics. Chem Rev 121:5158–5192. https://doi.org/10.1021/acs.chemrev.0c01213

Shetty AK, Shetty IN, Furtado ZV, Antony B, Boloor R (2012) Antibiogram of *Salmonella* isolates from blood with an emphasis on nalidixic acid and chloramphenicol susceptibility in a tertiary care hospital in coastal Karnataka: a prospective study. J Lab Physicians 4:74–077. https://doi.org/10.4103/0974-2727.105585

Shi W, Li K, Ji Y, Jiang Q, Wang Y, Shi M, Mi Z (2013) Carbapenem and cefoxitin resistance of *Klebsiella pneumoniae* strains associated with porin ompK36 loss and DHA-1 *β*-lactamase production. Braz J Microbiol 44:435–442. https://doi.org/10.1590/S1517-83822013000200015

Silbergeld EK, Graham J, Price LB (2008) Industrial food animal production, antimicrobial resistance, and human health. Annu Rev Public Health 29:151–169. https://doi.org/10.1146/annurev.publhealth.29.020907.090904

Soge OO, Adeniyi BA, Roberts MC (2006) New antibiotic resistance genes associated with CTX-M plasmids from uropathogenic Nigerian *Klebsiella pneumoniae*. J Antimicrob Chemother 58:1048–1053. https://doi.org/10.1093/jac/dkl370

Sousa MC (2019) New antibiotics target the outer membrane of bacteria. Nature 576:389–390. https://doi.org/10.1038/d41586-019-03730-x

Sun J, Li M, Lin M, Zhang B, Chen X (2021) High antibacterial activity and selectivity of the versatile polysulfoniums that combat drug resistance. Adv Mater. https://doi.org/10.1002/adma.202104402

Tambe S, Ochs MM, Hancock REW (2006) Role of the novel OprD family of porins in nutrient uptake in *Pseudomonas aeruginosa*. J Bacteriol Res 188:45–54. https://doi.org/10.1128/jb.188.1.45-54.2006

Threlfall J (2013) New research on antimicrobial resistance in foodborne pathogens. In: Sofos J (ed) In Woodhead publishing series in food science. Woodhead Publishing, Cambridge, pp 134–156

Ugboko H, De N (2014) Mechanisms of antibiotic resistance in *Salmonella* Typhi. Int J Curr Microbiol App Sci 3:461–476

Vergalli J, Bodrenko IV, Masi M, Moynié L, Acosta-Gutiérrez S, Naismith JH, Davin-Regli A, Ceccarelli M, van den Berg B, Winterhalter M, Pagès JM (2020) Porins and small-molecule translocation across the outer membrane of Gram-negative bacteria. Nat Rev Microbiol 18:164–176. https://doi.org/10.1038/s41579-019-0294-2

Viveiros M, Dupont M, Rodrigues L, Cout I, Davin-Regli A, Martins M et al (2007) Antibiotic stress, genetic response and altered permeability of *E. coli*. PLoS ONE 2(4):e365. https://doi.org/10.1371/journal.pone.0000365

WHO (World Health Organization) (2016) Burden of foodborne diseases in the South-East Asia Region. WHO, Regional Office for South-East Asia. https://www.who.int/publications/i/item/burden-of-foodborne-diseases-in-the-south-east-asia-region. Accessed on 26 July 2021

WHO (World Health Organization) (2017) The selection and use of essential medicines: report of the WHO expert committee. WHO technical report series. https://apps.who.int/iris/bitstream/handle/10665/259481/9789241210157-eng.pdf. Accessed ona 26 July 2021

Xu F, Ge C, Li S, Tang S, Wu X, Luo H, Deng X, Zhang G, Stevenson A, Baker RC (2021) Evaluation of nanopore sequencing technology to identify *Salmonella enterica* Choleraesuis var. Kunzendorf and Orion var. 15^+, 34^+. Int J Food Microbiol 346:109167. https://doi.org/10.1016/j.ijfoodmicro.2021.109167

Yang SK, Yusoff K, Thomas W, Akseer R, Alhosani MS, Abushelaibi A, Lim SHE, Lai KS (2020) Lavender essential oil induces oxidative stress which modifies the bacterial membrane permeability of carbapenemase producing *Klebsiella pneumoniae*. Sci Rep 10:1–14. https://doi.org/10.1038/s41598-019-55601-0